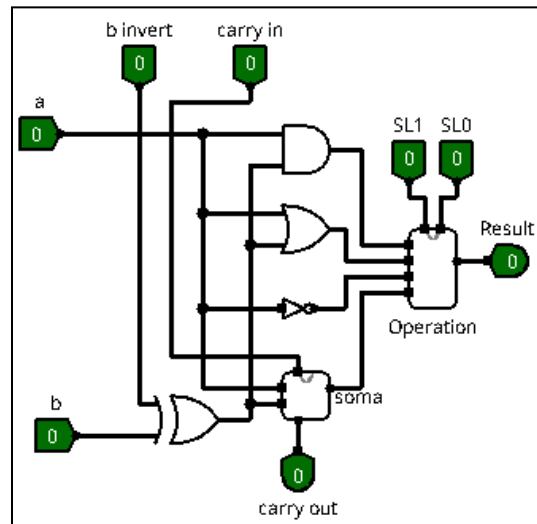


Aluno: Silas Hoffmann Reis Viana

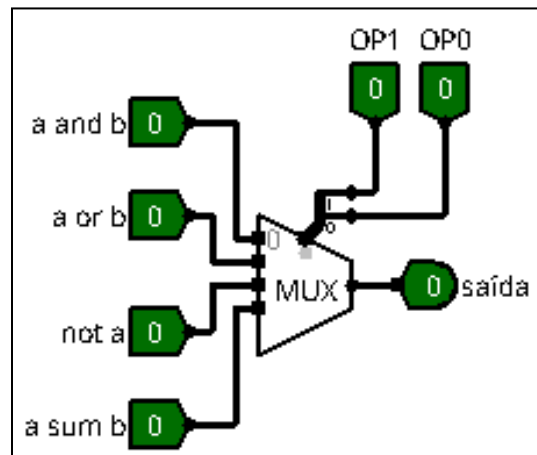
PARTE 1 → 6)

Instrução	Binário (A,B,OP)	Hexa	Resultado
AND(A,B)	0010 0001 00	(0000 1000 0100) = 0x084	0000
OR(A,B)	0010 0011 01	(0000 1000 1101) = 0x08D	0011
SOMA(A,B)	0010 0011 11	(0000 1000 1111) = 0x08F	0101
NOT(A)	1100 0000 10	(0011 0000 0010) = 0x302	0011
AND (B,A)	1100 1101 00	(0011 0011 0100) = 0x334	1100

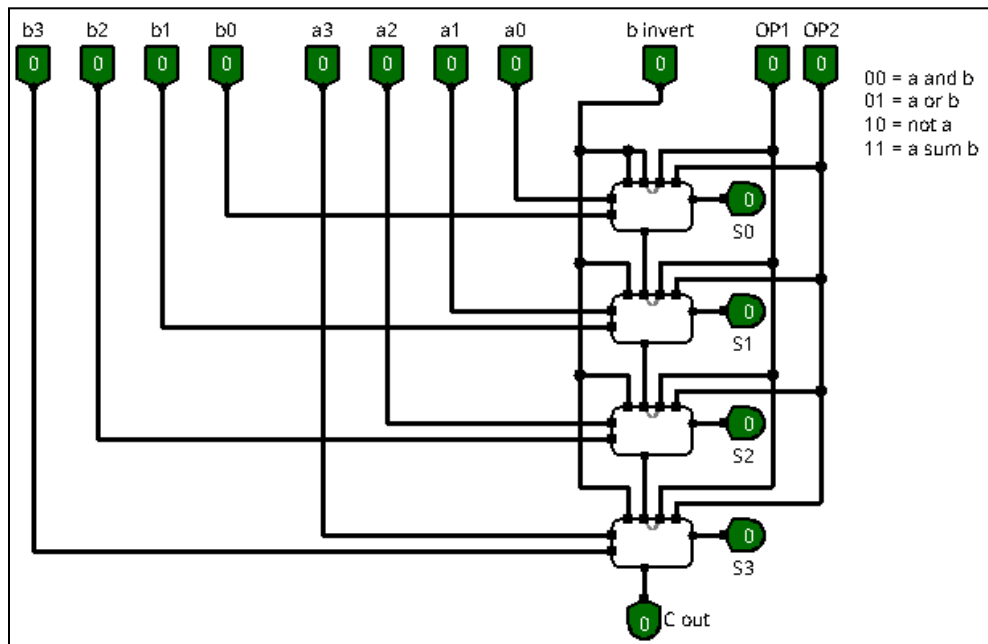
ULA



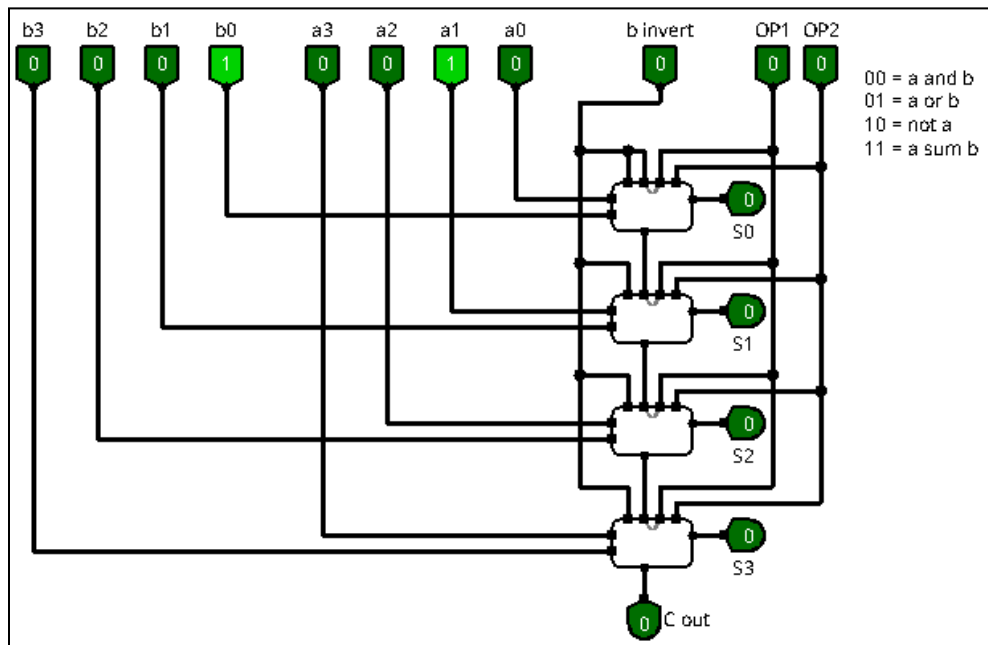
OPERATION



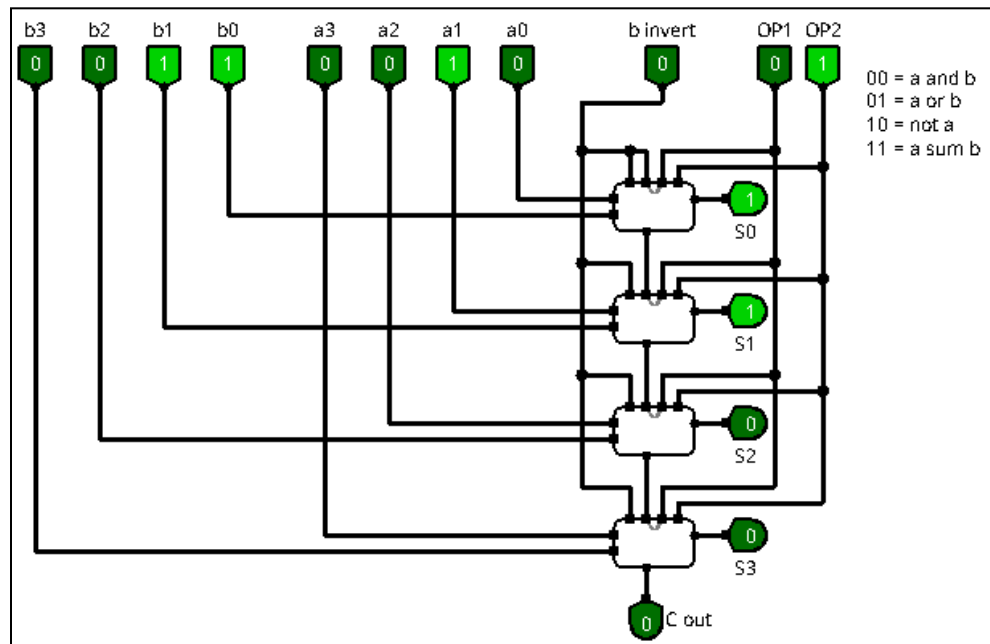
ULA 4 bits



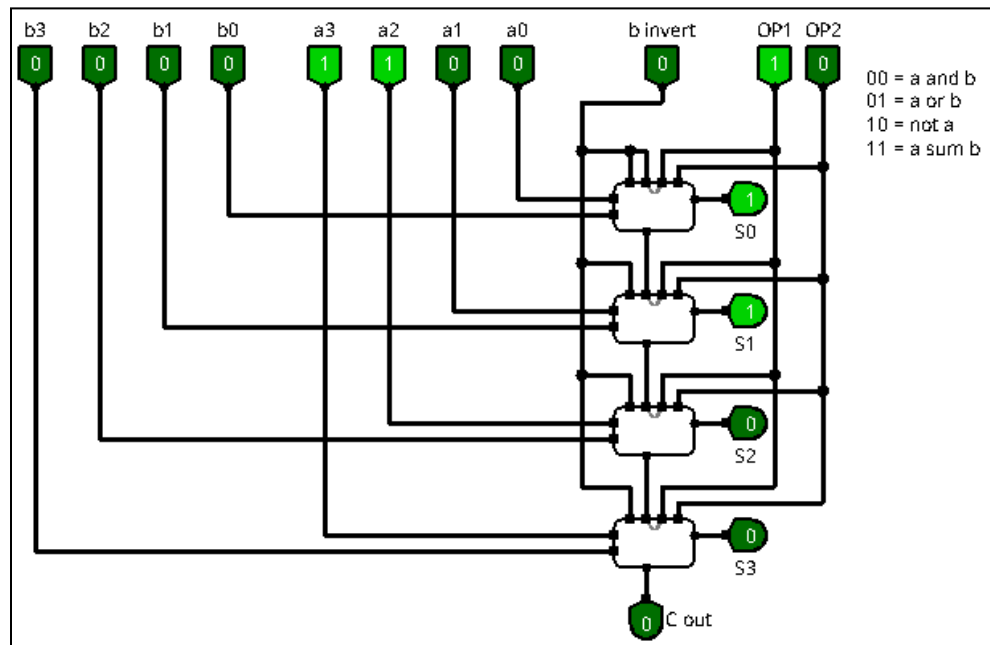
AND A B



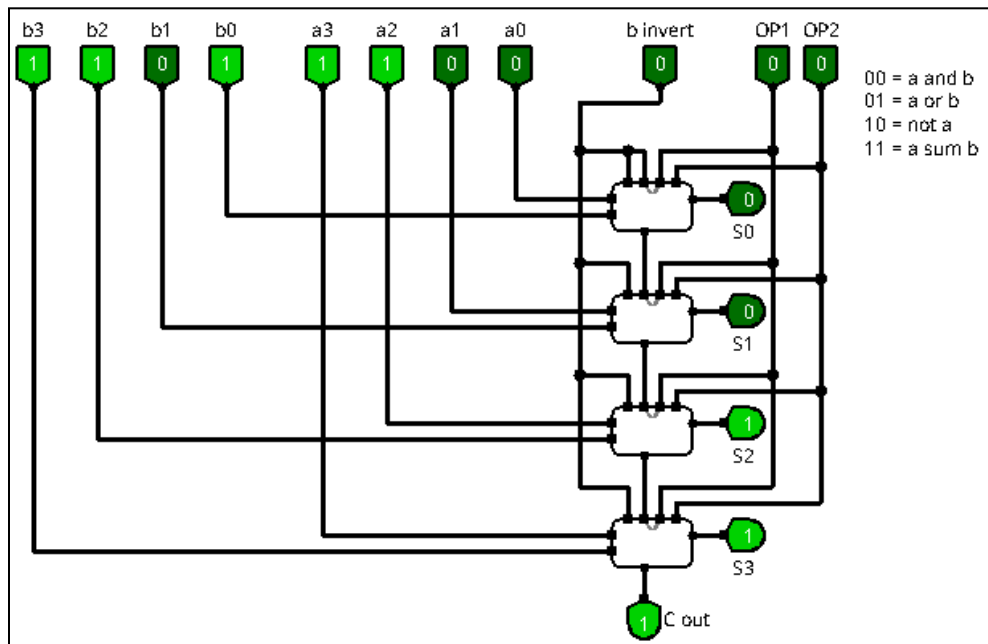
OR A B



NOT A

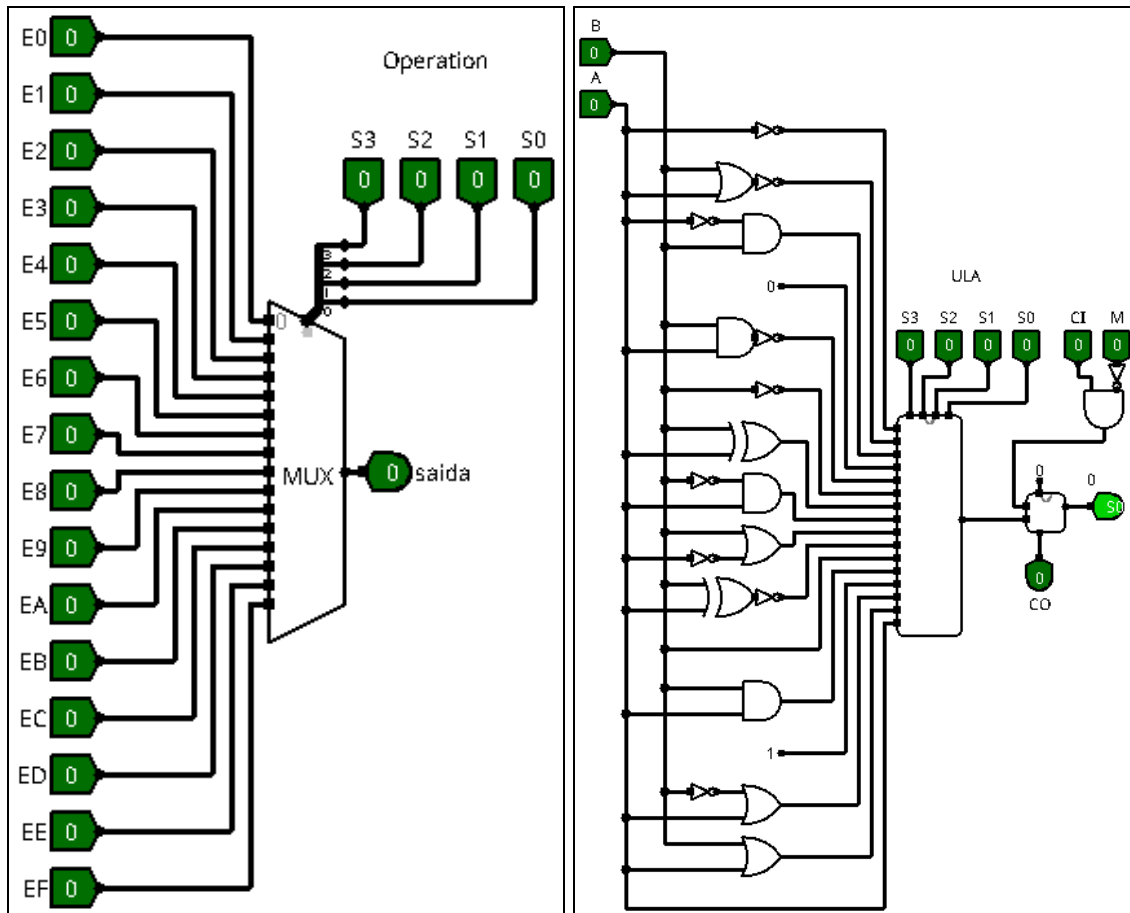


AND B A

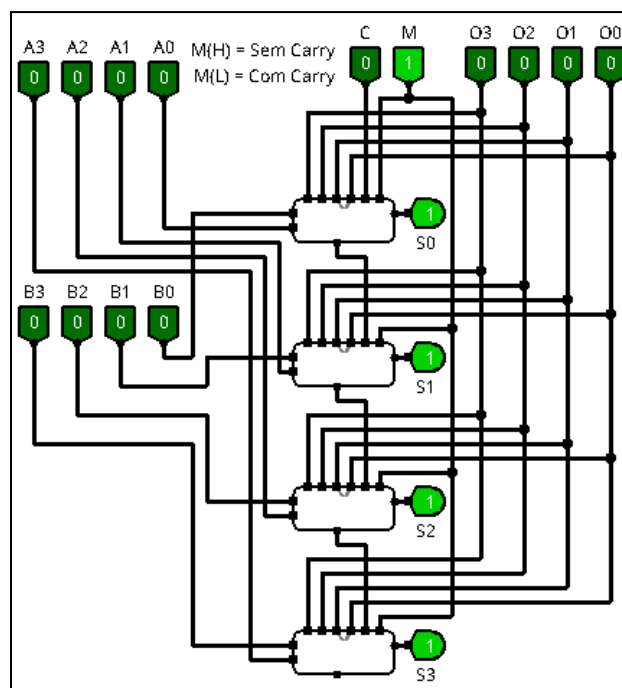


PARTE 2

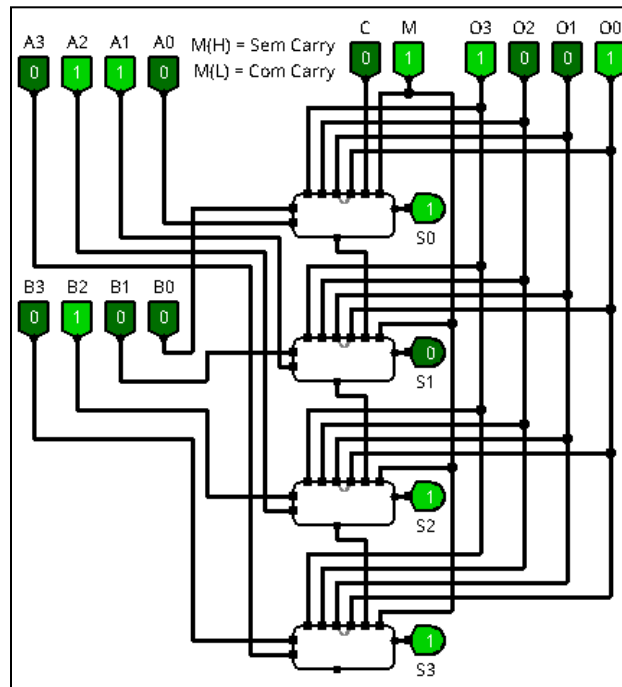
Instruções	Binário	Resultado da Operação
450	0100 0101 0000	B
CB1	1100 1011 0001	0
A32	1010 0011 0010	1
C43	1100 0100 0011	0
124	0001 0010 0100	F
785	0111 1000 0101	7
9B6	1001 1011 0110	2
CD7	1100 1101 0111	0
FE8	1111 1110 1000	E
649	0110 0100 1001	D
D9A	1101 1001 1010	9
FCB	1111 1100 1011	C
63C	0110 0011 1100	F
98D	1001 1000 1101	F
76E	0111 0110 1110	7
23F	0010 0011 1111	2



ULA 4 bits



Exemplo Prático (0010 0011 1111 → 0010)



Se o objetivo fosse realmente testar esta ULA, quantas linhas a nossa tabela verdade deveria ter, ou seja a tabela que você preencheu deveria ter quantas linhas?

Considerando que existem 16 possibilidades para o valor de A, B e Operation, englobar todas em uma tabela verdade precisará de $16 \times 16 \times 16 = 4096$ linhas, porém existem ainda o bit de Carry e Mode, quadruplicando as possibilidades, portanto serão necessárias 16384 linhas